

RESPIRATION — YEAST BREATHES TOO!

Planrs Academy | Grade 2 | Session 1 | 30 Minutes

■ DETAILED LESSON PLAN

Respiration — Yeast Breathes Too!

Subject	Grade	Duration	Session
Science — Life Processes	Grade 2 (Age 7–8)	30 Minutes	Session 1 of 1

BIG IDEA: "All living things breathe — even ones too small to see clearly, like yeast!"

ESSENTIAL QUESTION: "Does yeast breathe? How do we know?"

SECTION 1 — LEARNING OBJECTIVES

LO1	RECALL	State that all living things breathe (respire), including tiny organisms like yeast.
LO2	IDENTIFY	Identify yeast as a living organism that breathes and produces carbon dioxide gas (bubbles).
LO3	OBSERVE	Describe observations when yeast is mixed with warm water and sugar — bubbles forming, balloon inflating, bread-like smell.
LO4	CONNECT	Connect yeast breathing to Indian foods: idli batter rising, bread/pav puffing up, and dosa fermentation.
LO5	COMMUNICATE	Use simple words to explain: "Yeast is alive. It breathes. It makes bubbles. That is why idli and bread are fluffy!"

SECTION 2 — KEY CONCEPTS WITH INDIAN EXAMPLES

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■ CONCEPT 1: Breathing / Respiration

Breathing is something all living things do to get energy and stay alive. When we breathe, we take in air and give out another gas.

- Breathing helps us run, play, think, and grow.
- Every living thing — human, dog, fish, plant — breathes in its own way.
- Yeast breathes too, just quietly and invisibly!

■■ Indian Example: Just like you breathe harder when playing kabaddi or kho-kho in your school ground, yeast also breathes — in its own tiny, quiet way.

■ CONCEPT 2: Yeast — A Living Thing

Yeast is a very tiny living thing — so small we cannot see it with our eyes alone. It is neither a plant nor an animal. But it IS alive!

- Yeast comes in small packets (like Bluebird brand) available at any kirana store.
- It looks like a dry, yellowish powder when you open the packet.
- It becomes active (wakes up!) when given warm water and food (sugar).

■■ Indian Example: At the neighbourhood bakery or Irani café, the baker adds yeast to dough to make soft pav and bread. The yeast wakes up and starts breathing in the warm dough!

■ CONCEPT 3: Carbon Dioxide — The Gas Yeast Breathes Out

When yeast eats sugar and breathes, it breathes out a gas called carbon dioxide. We can SEE this gas as bubbles!

- Carbon dioxide is the same gas WE breathe out too.
- In our experiment, this gas has nowhere to go — so it fills the balloon!
- Bubbles = proof that yeast is alive and breathing.

■■ Indian Example: Have you seen bubbles in your chaas (buttermilk) or lassi? Have you noticed idli batter bubbling overnight? That's CO₂ made by tiny living things breathing!

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■ CONCEPT 4: Sugar as Food for Yeast

Yeast needs food to breathe and make energy. Its favourite food is SUGAR! When we give yeast sugar and warm water, it becomes active.

- Without sugar, yeast stays 'asleep' — no bubbles, no gas.
- Add sugar → yeast wakes up → yeast breathes → bubbles appear!
- Warm water is important — cold water keeps yeast asleep.

■■ Indian Example: Just like you need dal-chawal or roti-sabzi to have energy to play, yeast needs cheeni (sugar) to have energy to breathe.

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SECTION 3 — SESSION OUTLINE WITH TIMING (30 MINUTES)

0–3 min

HOOK: The Mystery Bag

- Show students a small packet of dry yeast.
- Let them smell it safely — ask: 'Does this smell like something from home?'
- Ask: 'What do you think is in this packet? Is it alive?'
- Build curiosity: 'This is YEAST. Let's find out if it breathes!'

■ NOTE

Materials needed: 1 yeast packet, curiosity!

3–7 min

CONCEPT INTRO: What is Breathing?

- Ask students to take a big breath in and out together.
- Discuss: 'We breathe to stay alive and get energy.'
- Show the visual chart: Human → Breathes! Dog → Breathes! Fish → Breathes! Plant → Breathes! Yeast → ???
- Ask: 'Do you think yeast breathes? Let's find out!'

■ NOTE

Slide reference: Slides 3–4 from the deck

7–18 min

EXPERIMENT: Wake Up, Yeast!

- Step 1 — Pour warm water into bottle (warm like chai, NOT boiling).
- Step 2 — Add 1 tsp yeast into bottle. Watch it fall in!
- Step 3 — Add 1 tsp sugar: 'This is yeast's food!'
- Step 4 — Stretch balloon tightly over the bottle mouth.
- Step 5 — Wait and WATCH. Ask: 'What do you see? What do you smell?'
- Observe: Bubbles form, mixture gets frothy, balloon slowly inflates.
- Explain as it happens: 'Yeast ate the sugar. It breathed. It made gas. Gas = bubbles = balloon filling up!'

■ NOTE

Safety note: Use warm (not hot) water. Teacher handles water pouring.

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18–23 min

INDIAN KITCHEN SCIENCE CONNECTION

- Show images/real items: idli batter bowl, bread/pav, dosa batter.
- Ask: 'Have you seen idli batter bubble the next morning?'
- Explain: 'Tiny living things (yeast's cousins!) breathed all night in the batter.'
- 'That CO₂ gas made pockets in the batter — and when steamed, we get fluffy idli!'
- Connect to pav: 'Bakery uncle adds yeast to dough. Yeast breathes, gas fills dough, pav becomes soft!'

■ NOTE

Slide reference: Slides 10–11 from the deck

23–27 min

RECAP: Say It With Me!

- Call-and-response: Teacher asks, students answer loudly:
- → 'Yeast is...?' → ALIVE!
- → 'Yeast breathes and makes...?' → BUBBLES!
- → 'The gas yeast makes is...?' → Carbon dioxide!
- → 'That's why idli is...?' → FLUFFY!
- Ask 2–3 students to explain in their own words what they saw.

■ NOTE

Assessment moment: note student understanding here

27–30 min

CLOSE & HOME ACTIVITY

- Send students home with a mission: 'Ask your mummy or nani — are you making idli batter tonight?'
- 'Watch if it bubbles. Now you know WHY!'
- Celebrate: 'Today YOU are real scientists! You proved yeast breathes!'
- Distribute worksheet if using (optional, can be homework).

■ NOTE

Take-home message: Yeast is alive. It breathes. That's why our food is fluffy!

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SECTION 4 — EXPERIMENT: WAKE UP, YEAST! (FULL GUIDE)

Materials Required (Per Group / Demo)

Material	Where to Buy (India)	Approx. Cost
Dry yeast packet (Bluebird / Gloripan)	Kirana store / Big Bazaar / D-Mart	■10–■20
Sugar (cheeni)	Kirana store	■5
Warm water	School kitchen / tap	Free
Small plastic bottle (500 ml)	Provision store	■10–■15
Balloon (small)	Stationery / toy shop	■5–■10
Measuring spoon (1 tsp)	Kitchen / stationery	■5–■10

Step-by-Step Procedure

Step 1	Pour Warm Water Pour warm water into the bottle — about half full. The water should feel warm on your wrist, like a comfortable bath. NOT hot or boiling! ■ <i>Why warm? Cold water keeps yeast 'asleep'. Warm water wakes it up!</i>
Step 2	Add Yeast Open the yeast packet and pour it into the bottle. Watch the tiny yeast grains fall into the warm water. They look like tiny dots! ■ <i>Fun to say: 'Wake up, little yeast! Time to get to work!'</i>
Step 3	Add Sugar — Yeast's Food! Add 1 teaspoon of sugar into the bottle. This is YEAST'S FOOD! Sugar gives yeast energy to breathe. Swirl the bottle gently to mix. ■ <i>Without sugar, nothing happens. Sugar is the magic ingredient!</i>
Step 4	Put the Balloon On Stretch the mouth of a balloon over the opening of the bottle. Make sure it fits tightly so no gas escapes out the sides. ■ <i>Teacher tip: Pre-stretch the balloon a bit before class to make it easier for students.</i>

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Step 5

Wait and Watch!

Place the bottle where everyone can see it. Now WAIT and OBSERVE! After 5–10 minutes, you will see bubbles in the liquid and the balloon will start to inflate!

■ *Observation prompts: What do you see? What do you smell? Is the balloon changing?*

■ What Happens & Why

- Yeast eats the sugar for energy — just like we eat food for energy.
- As yeast digests sugar, it produces carbon dioxide gas as a by-product.
- The gas bubbles up through the liquid (frothy mixture).
- Gas rises into the balloon and inflates it — PROOF that yeast is breathing!
- You may also smell a warm, bread-like or yeasty aroma — that's yeast at work!

■■ Troubleshooting — What If It Doesn't Work?

Balloon not inflating? → Check water isn't too cold. Rewarm slightly and try again.

No bubbles? → Add a little more sugar. Old yeast may be expired — check the packet date.

Balloon popped off? → Ensure it's stretched tightly. Use a fresh balloon.

No smell? → Normal — smell varies by yeast brand. The visual evidence (bubbles) is enough.

No materials available? → Show a video demonstration. Never skip the concept!

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SECTION 5 — INDIAN KITCHEN SCIENCE CONNECTIONS

■ Idli & Dosa Batter

Idli and dosa batter is made from soaked rice and lentils, then left overnight to ferment. Tiny living things (bacteria and yeast-like organisms) breathe in the batter all night long — releasing carbon dioxide bubbles. By morning, the batter is puffed up, bubbly, sour-smelling, and ready! When steamed, the gas pockets become the soft holes inside each idli.

■■■ *Teacher Ask: Ask: 'Have you ever seen your mummy's idli batter the next morning? Was it bubbly? Did it rise up in the bowl?'*

■ Pav / Bread from the Bakery

The 'Bakery Uncle' adds yeast to flour dough when making pav, bread, or naan. As the dough rests (called 'proofing'), yeast breathes and releases carbon dioxide. The gas gets trapped in the sticky dough, making it puff up like a balloon! When baked in the oven, the gas bubbles set — giving us soft, airy bread.

■■■ *Teacher Ask: Ask: 'Have you seen dough 'rise' before baking? That's yeast breathing and making gas bubbles inside!'*

■ Jalebi Batter

In North India, jalebi batter is made from maida (refined flour) and left to ferment for a day. Yeast and bacteria breathe in it, making the batter slightly sour and bubbly. When fried, it gives jalebi its signature crispness and slight tang.

■■■ *Teacher Ask: Ask: 'Have you had jalebi that tastes a little tangy? That's the tiny living things who breathed in the batter!'*

■ Dahi (Curd) — A Parallel!

Though dahi uses bacteria (not yeast), the idea is the same: tiny living things are ALIVE in the curd starter. When added to warm milk, they breathe, multiply, and change the milk into thick, sour dahi by morning. A great analogy for students who aren't familiar with idli.

■■■ *Teacher Ask: Say: 'Just like the tiny things in curd starter turn milk into dahi overnight, yeast turns batter into fluffy idli!'*

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SECTION 6 — FORMATIVE ASSESSMENT (5 QUESTIONS)

Instructions: Complete the following. Circle your answers. Draw where asked. No wrong answers — just try your best! ■

Q1

[MCQ]

Yeast is a:

A) Rock B) Tiny living thing ✓ C) Type of water D) Metal

✓ ANS

Answer: B — Yeast is a tiny living thing.

Q2

[MCQ]

What gas does yeast make when it breathes?

A) Oxygen B) Steam C) Carbon dioxide ✓ D) Smoke

✓ ANS

Answer: C — Yeast makes carbon dioxide gas.

Q3

[True / False]

Only humans and animals breathe. Yeast does NOT breathe.

True False ✓

✓ ANS

Answer: False — Yeast breathes too!

Q4

[True / False]

Yeast needs sugar and warm water to wake up and breathe.

True ✓ False

✓ ANS

Answer: True — Yeast needs sugar (food) and warm water to become active.

Q5

[Draw & Label]

Draw and label the experiment! Draw: 1) The bottle with yeast, water and sugar
2) The balloon getting bigger 3) Bubbles inside the bottle

[Drawing space — provide lined/blank space on student copy]

✓ ANS

Expected: A bottle, balloon inflating on top, bubbles inside the liquid.

SECTION 7 — RECAP CHANT & SUCCESS CRITERIA

■ Call-and-Response Recap

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TEACHER SAYS...	STUDENTS SAY...
Yeast is...?	■ ALIVE!
Yeast breathes and makes...?	■ BUBBLES!
The gas yeast makes is called...?	■ Carbon dioxide!
That's why our idli is...?	■ FLUFFY!
Today YOU are...?	■ REAL SCIENTISTS!

■ Success Criteria

Student Success Indicators

- ✓ 80%+ students correctly state yeast is a living thing.
- ✓ 80%+ students say yeast breathes and makes bubbles.
- ✓ Students connect to at least ONE Indian food example.
- ✓ Students show visible enthusiasm during the experiment.
- ✓ Worksheet completed with 70%+ accuracy.

Content Quality Success Indicators

- ✓ All Indian context references are culturally accurate.
- ✓ Language is Grade 2 appropriate (Flesch-Kincaid ≤ 2.5).
- ✓ Experiment replicable with materials under ₹60 total.
- ✓ No instances of Western-default cultural references.

Curriculum Alignment Success Indicators

- ✓ Aligns with NEP 2020 Foundational Stage: curiosity, hands-on learning.
- ✓ References NCERT EVS themes for Class 2: Living and Non-Living Things.
- ✓ Reviewed by subject matter expert + Grade 2 classroom teacher.

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■ **Take-Home Activity:** Tonight, ask your family: "Are you making idli or bread batter?" Watch if it bubbles before it's ready. Now YOU know why — tiny yeast are breathing and making bubbles! You are a real scientist! ■

"Every child is a scientist waiting to discover the world." — Planrs Academy